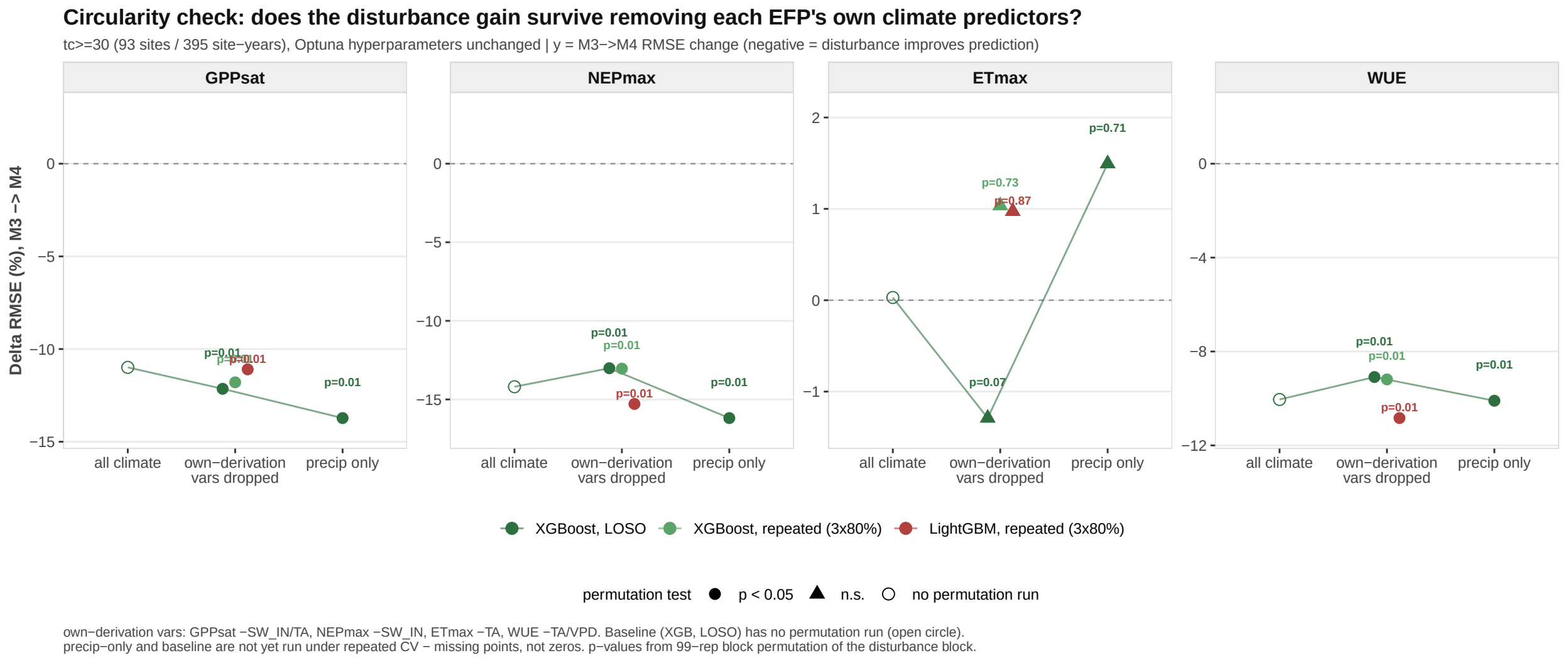
Circularity check: does the disturbance gain survive removing each EFP's own climate predictors?
tc>=30 (93 sites / 395 site−years), Optuna hyperparameters unchanged | y = M3−>M4 RMSE change (negative = disturbance improves prediction)
GPPsat
NEPmax
ETmax
WUE
Delta RMSE (%), M3 −> M4
all climate
own−derivation vars dropped
precip only
p=0.01
p=0.71
p=0.73
p=0.87
p=0.07
XGBoost, LOSO
XGBoost, repeated (3x80%)
LightGBM, repeated (3x80%)
permutation test
p < 0.05
n.s.
no permutation run
own−derivation vars: GPPsat −SW_IN/TA, NEPmax −SW_IN, ETmax −TA, WUE −TA/VPD. Baseline (XGB, LOSO) has no permutation run (open circle).
precip−only and baseline are not yet run under repeated CV − missing points, not zeros. p−values from 99−rep block permutation of the disturbance block.